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Problem Chosen

B**2007 Mathematical Contest in Modeling (MCM) Summary Sheet**

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Summary

Airlines desire to reduce the turnaround time, which is affected mainly by boarding time. We propose a new boarding procedure to minimize the boarding time.

We find two types of interferences: aisle interference and seat interference. The interferences prolong the boarding time. Three principles are put forward to reduce interferences: "Window first, middle followed, aisle last"; "Passengers with seats in the same row should not boarding in one group"; "Farther from the door, earlier boarding".

Based on the above principles, the boarding time is sum of two parts. One is the sum of groups' boarding time. The other is the sum of time interval between two adjacent groups.

Considering one group, the average boarding time depends on the quantity of passengers in the group. Assuming average baggage time is 8 seconds, we compute the expectation of average boarding time. And the expected average boarding time is minimized, if and only if the number of passengers in one group is 12.

The time interval between two adjacent groups is the time that first passenger of the latter group walks from cabin door to the assigned seats zone.

Our "single column filling in groups" procedure can be applied to all small airplanes. It can be realized easily by gate agents announcing a particular group boarding a assigned area. Without any loss of generality, the procedure is also valid for midsize and large planes, by dividing the plane into areas according to the aisles and doors.

Airbus A320 is chosen as a typical small airplane, then, boarding procedure is devised: passengers are arranged into 12 groups, with 12 or 13 passengers each group, and boarding in order. We use discrete event system simulation, under the condition that the baggage time and time passing one row obey normal distribution (respectively, means 8 and 2.8, standard deviations 1 and 1/3).

We obtain the boarding time for small plane (A320, 22.6 min), midsize (Boeing767, 30.6 min), large (A380, 50.5 min). Especially for small plane, compared with the "out-side-in", "reverse pyramid", and "from back to front", our procedure improves the boarding efficiency by 8.6%, 13.9%, 17.4%.

The boarding time for each group is 1.9 min, which is appropriate for both passengers and gate agents. Therefore, our procedure is practical. The boarding time is insensitive to baggage time, however, relatively sensitive to time passing one row.

At last, we recommend our procedure to airline executives, and set out that the procedure is economical, practical.

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Each team member must sign the statement below:

(Failure to obtain signatures from each team member will result in disqualification of the entire team.)

Each of us hereby testifies that our team abided by all of the contest's rules and did not consult with anyone who was not on this team in developing the enclosed solution paper.

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Mailing Address and Signature of Zeng Xianwen

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An Analysis of the Kidney Transplant Network

An Analysis of the Kidney Transplant Network

Introduction

Since the first case of the kidney transplant surgery proclaimed a success in early 1950's between the twins, organ transplant has saved a large amount of lives. Today, most organ transplants are relatively safe, routine procedures. Now transplantation is considered to be the best treatment option for thousands of patients every year, among which kidney transplant plays an important role.

Unfortunately, doctors and patients always face an obstacle: the demand for transplants has far surpassed the supply of donated organs. Patients must wait months, even years, for their chance at recovery. In United States, the average waiting time exceeds three years-double that in some areas.

Long waiting list makes failures in saving the lives of desperate need candidates. The problems raised are how to model the generic US transplant network and make improvement on it. As to the kidney transplant, detail issues such as ways to maximize the exchange system and strategy for patients are left to be discussed.

Our work focuses on the optimization of organ match and the effectiveness of network system. The purpose is to make full management of the matching process and the post-graft survival process. What we should do are as following:

- a) Model the network and give insights into bottlenecks, efficiency improving, etc.
- b) Investigate other nation's transplant policies and give recommendations on the improvements of the US transplant network;
- c) For kidney transplant, devise a procedure to maximize the number and quality of the exchange system;
- d) Devise a strategy for a patient to decide whether to take an offered kidney;
- e) Analyze the ethical issues and recommend changes on criteria/policies;
- f) Analyze the potential risks of donors and suggest ways to recruit more donors.

Task1: The Genetic Model of US Transplant Network

Symbols

Table 1. Symbol Table

Symbol	Meaning
P	The probability of organs being successfully utilized
D_i	The amount of organ donation in match way i
	The probability of successfully matching in match way i
m_i	The amount of organs discarded in match way i
	The survival probability of transplant in match way i
d_i	The total number of match ways
	The total number of patients

Model the Transplant Network

Assumptions

- The number of organ donation is far less than that of patients in need
- The rate of people with different blood type in patient is the same as the donors
- The blood type match and HLA match is independent
- the matching process and transplant process are independent

Analysis on the Network

In this task, we are required to build a mathematical model for the generic US transplant network. From the view of the Organ Procurement Transplant Network (OPTN), our work focuses on the efficiency of the network system.

The organ transplant network mainly includes two modules:

a) Matching process-matches donated organs to patients with organ needs. This process is consisted of tests, such as blood type, tissue type, size of the organ, time on the waiting list, distance between donor and recipient, etc;

b) Post graft survival process-implements the organ transplant surgery. This process describes the potential risks and survival time for an organ transplant.

Besides, there is another factor in the system-organ discard. Organ discard takes place when the donor is willing to donate his organ, and his organ information is given to OPO for match. But for some unpredicted reason, when the organ is really donated, it is already impossible to survive and has to be discarded. This part doesn't go to surgery, so the realistic organs can be transplanted is less than matched pairs.

We analyze the transplant network by taking it as a system shown in **Figure 1**:

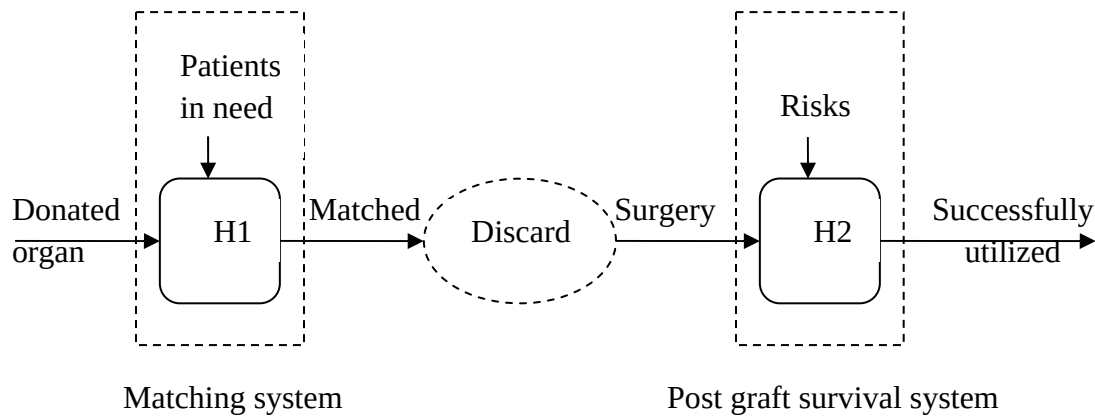


Figure 1. A basic transplant network system

The amount of donated organs is the system input, and the number of organs which are successfully utilized is the system output. The matching process and post-graft survival can also be seen as two subsystems: matching-system and post graft survival-system. Based on the assumption that the matching process and transplant process are independent, the organ transplant procedure can be identified as a linear system. The line between the subsystems is transmitting-channel, and the number of discarded organs is the channel cost. The risks include the failure of surgery, the degradation of organ after transplant but before the expected survival time and so on. The symbols H1, H2 stands for system functions.

For the matching system, we can divide it into three parts. Assume blood type and HLA markers are necessary tests for all, but different transplant types may lead to different additional tests. Considered the general applicability of our model, we reckon the extra tests relative to transplant type as other tests. And for post graft survival system, we divide it into two parts: surgery and survival time. Then the former system can be refined as **Figure 2**:

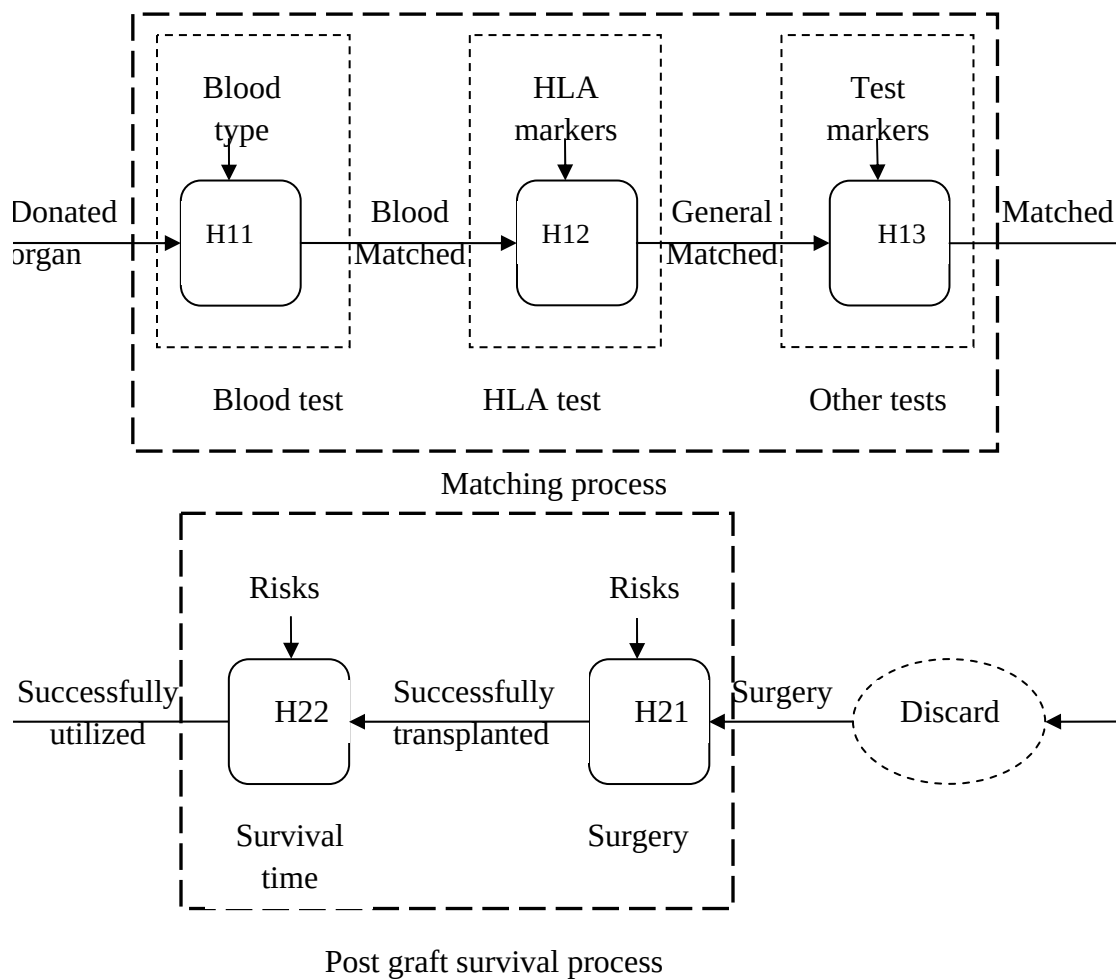


Figure 2. A more accurate transplant network system

In the transplant system, we put forward the concept of match way, which means if we want to match an organ to patient in need, we have to test the physical distance between the donated organ and recipient, including blood type, HLA markers, etc. Then match ways are determined by the factors in the tests. If they are compatible, we say in this way they are a match pair. For example, a donated organ requires that the blood type A, zero-mismatches and with no additional desires for compatibleness. Then blood type A, zero-mismatches is a match way.

Since the number of donors is far less than that of patients in need, our optimum goal is to increase the successful transplant rate of donated organs, from which we increase the ultimate system efficiency of the transplant network. So the process of choosing candidate according to priorities on the waiting list can be left out, for this process embodies the personal benefit, but the purpose of our model is to try to make effective use of donated organs. This means we only want to find a candidate to take organ transplant no matter what his status is on the waiting list.

Generic model of Transplant network

Based on the analysis of the system above, we develop a generic model of transplant network as below:

$$P_u = \frac{\sum_{i=1}^n (D_i m_i - d_i) \cdot r_i}{\sum_{i=1}^n D_i}, \quad (1)$$

where P_u is the probability of organs being successfully utilized, n is the total number of match ways, D_i is the donated organs in match way i , m_i is the probability of successfully matching in match way i , and it is affected by the total amount of candidates, the blood type, HLA markers, the type of organ transplant, etc. When D_i is put into the system with a successful match probability m_i , the match system's output is $D_i m_i$, which means the total matched pairs of patients in match way i .

And d_i is the amount of organs discarded and it is mainly influenced by the terminal conditions of donors. The term $D_i m_i - d_i$ means the pairs who can implement the transplant surgery in match way i , r_i is the successfully utilized rate of the surgery in match way i , which means the organ is successfully transplanted and can survival for a certain time, such as five years. The ultimate system output is $(D_i m_i - d_i) \cdot r_i$, which represents the number of patients saved in match way i .

Notice: To an organ, there may be more than one match ways, but the organ can only be matched with one patient. So for a certain organ in different match ways, it can only enter a certain subset of D_i .

Run the model

Because the number of the match ways differs when the organs are different, to analyze the problem in detail, we take kidney transplant as an example to get the numerical solution.

From the US transplant policies that for kidney transplant, the matching requires identical blood type zero antigen mismatched candidates. And we suppose no additional tests are taken. In this case, there is only one match way, namely, $n=1$.

So the generic model can be simplified:

$$P_u = \frac{(Dm - d) \cdot r}{D} \quad (2)$$

Where D is the total kidney donation, m is the matching probability. d is the kidney discarded and r is the survival rate.

Estimate the match probability in the matching process

If a patient wants to accept the kidney of a donor, the transplanted kidney must be of a compatible ABO blood type and no more than three mismatches on six HLA marks in the blood. Based on the assumption that the blood type match and HLA match are independent, we can compute the match

probability by multiplying the HLA match rate and the ABO blood type match rate.

For the HLA match rate is 1/4000 [Ref.1], we only need to determine the blood type match rate:

The ABO blood type can be matched as shown in **Figure 3** below:

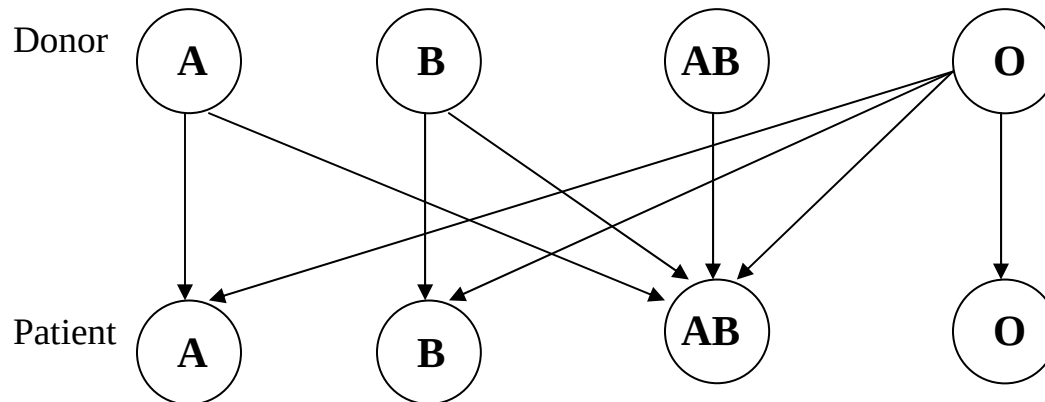


Figure 3. The ABO blood type match

The recent number of donors by different ABO blood type is shown in **Table 2** (Data source: OPTN, 2007):

Table 2. The number of donors by Different ABO blood type

Blood type	Number of donor
All ABO	101,634
A	30,595
B	15,283
AB	2,827
O	52,929

We can approximate the rate of ABO type is AB: B: A: O=1:5:10:18. If a donor with a blood type A, the probability that a discretionary patient can be match with him by blood type is $\frac{10}{34} + \frac{1}{34} = \frac{11}{34}$. As the same reason, if a donor with a blood type B, O or AB, the probability that a discretionary patient can be match with him by blood type is $\frac{7}{34}, \frac{1}{34}$ or 1. The expected matching probability for an organ with a patient is:

$$p = \left(\frac{18}{34} \times 1 + \frac{10}{34} \times \frac{11}{34} + \frac{5}{34} \times \frac{6}{34} + \frac{1}{34} \times \frac{1}{34} \right) \times \frac{1}{4000} = 1.62 \times 10^{-4}$$

Considering there are some other factors may reduce the match probability, we let the match probability be 10^{-4} for convenience.

If an organ is put into the system, the total candidates are put into the system for matching, which can be seen as independent repeated experiments. The successful matching means we find at least one candidate who fit the organ transplant. So using the method of probability, we can get the successfully matching probability is:

$$m = 1 - (1 - p)^w$$

Where w is the number of patients on the waiting list. When the w is very large, the successful-

matching probability will approach 1. In recent situation, w is large enough thus we can assume that in kidney transplant, the probability of matching is $m=1$.

The donated organs come from three ways: a) Standard Criteria Donor (SCD); b) Expanded Criteria Donor (ECD); c) Donation after Cardiac Death (DCD).

The total amount of donated kidneys in 2004 is 12972 (OPTN, 2007). The statistic data of discarded organs in 2004 is shown in **Table 3** (OPTN, 2007):

Table 3. The organs discarded in United States in 2004

Donor type	SCD	ECD	DCD	Total
Organs discarded	835	1077	171	2083

From the table, we know the discarded organ is 2083 per year. And we can find the survival rate of kidney transplant with zero-mismatches and average survival time of 5 years is 77.1% (OPTN, 2007). Now substitute these statistic data into Equ.2, we can get the probability of successfully transplanted organ in 2004 is: $P_u = 0.6455$.

Insights into the Model

Potential bottlenecks

From the model developed, we can see that three variables in organ transplant lead to the change of the probability of successfully transplanting: the amount of organs donated D_i in match way i , the discarded organs d_i and the survival rate r_i . Any change in them may lead to potential bottlenecks for efficient organ matching.

Procurement bottleneck

Recently, the shortage of donated organs has already decreased the effectiveness of US transplant network. The large requirement but relatively small donation leads to a long waiting list. Many failures come out because no suitable organ offered to them.

For instance, we analyze the change rate of patients in waiting and kidney donation, and the result indicates that this shortage may strengthened and become a bottleneck of US transplant network.

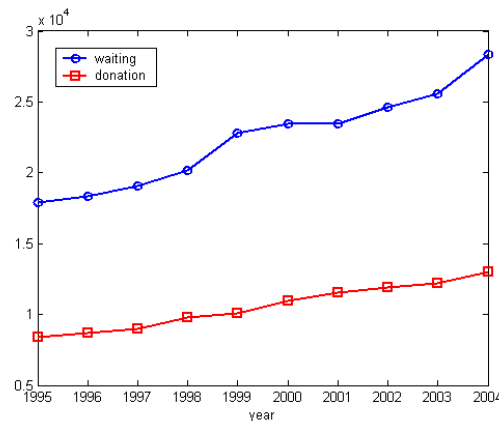
Table 4 shows the historical data of patients on the waiting list and the organ donation, and using MATLAB, we plot the data shown in **Figure 4**.

Use MATLAB to plot the trend of data along time. From the figure we can see that both donation and waiting follow linear increase. Using SPSS for linear regression, we get the correlation coefficient between donation and time is 0.9925, meanwhile the correlation coefficient between waiting patients and time is 0.9606. This means there is a linear correlation between donation and time. And there is also a linear correlation between number of patients on the waiting list and time.

Using linear regression, we can get the increase rate of donation is 608 per year and the waiting increase rate is 1602 per year. The large difference in the rate of increment

Table 4. The number of kidney donation and waiting (1995-2004)

year	Number of waiting list	Number of donation
1995	17875	8393
1996	18328	8714
1997	19047	9016
1998	20171	9760
1999	22800	10110
2000	23427	10982
2001	23445	11566
2002	24636	11878
2003	25604	12226
2004	28320	12972

**Figure 4.** The number of kidney donation and waiting (1995-2004)

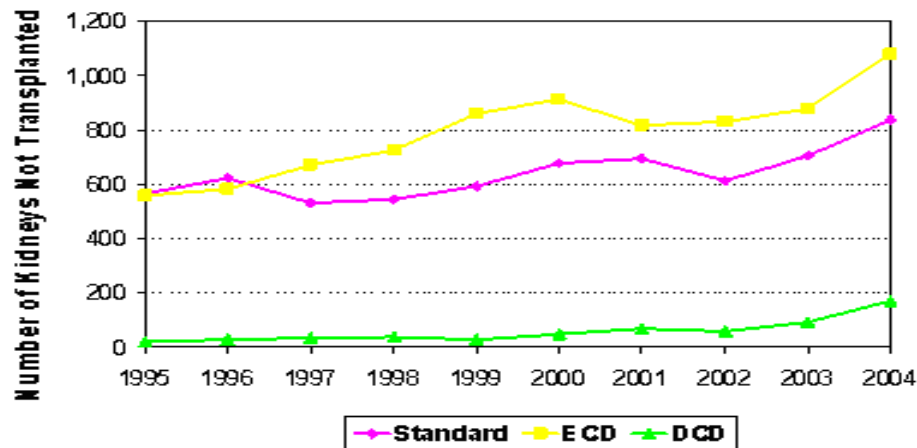
means the

gap between waiting and donation is on a rise. If the situation goes on, there will certainly be a bottleneck in organ procurement.

Discarding bottleneck

The effective utilization rate of an organ is relatively low, why? The discard is responsible. Even though a donor consents to donating, his organ may not be used for some discouraging news, such as the degradation on health. This bad news put a matched-patient into failure. Furthermore, this is a significant loss to the donor population and become a potential bottleneck.

With the data of organ discarded from 1995-2004 (data source: OPTN, 2007), the raising trend is shown in **Figure 5**:

**Figure 5.** Number of kidney discards by donor type (1995-2004)

The total number discarded is the sum of three donor types. To analyze the influence of the organ discard, we focus on the average percentage of discarding per year. The following **Figure 6** shows the kidney discard rate in the total donation from 1995 to 2004:

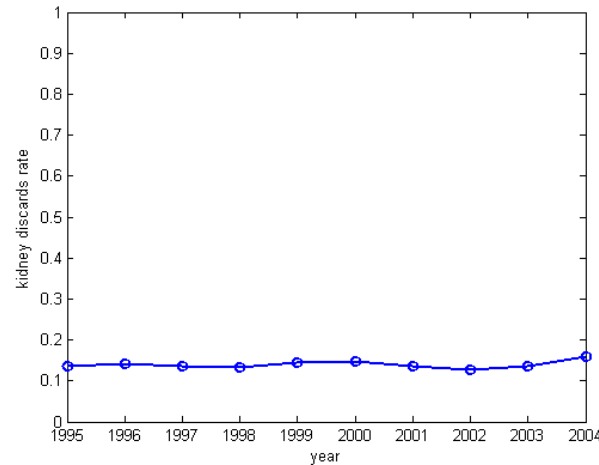


Figure 6. Kidney discard rate (1995-2004)

From the figure we can see that the kidney discard rate is in a steady state and the average rate is 14.04%. This is quite a large portion and makes the transplant network less efficient and the loss may discourage the organ donors.

Technical bottleneck

Technique bottleneck is caused by the survival rate r_i of post graft survival process. For example, in recent level of medical treatment, kidney transplant has a survival rate of 0.771 for zero-mismatches five-year survival time, and the data is 0.750 for one-mismatch five-year survival time. For this reason, parts of the patients can't recover even if they take transplant surgeries successfully. Some side effects will worsen the health of them, and the life expectancy will be shortened. The advanced medical technology can certainly make efficient use of donated organs and save more patients under the same level of donation.

Other predictable bottlenecks

Financial bottleneck: To take an organ transplant is of high expenditure, and many patients can't afford it. Even if they can pay the transplant cost, the medical treatment after transplant is far more than imagination. So many patients give up the chance to recover. And the rich ones in low priority take the surgery. This brings the financial bottleneck.

Conception bottleneck: We know that the sources of donation are deceased and live donation. Everyday, a lot of people deceased, but they do not donate their organs to the people in need. And to the ones alive, only a few people sign their name on the donor sheet. Why this happens? Someone says people don't want their reliquiae shattered for it disobeys traditional convention. And live donation occurs most in relatives. People are unwilling to help others because of the potential risks. They say that organ donation is a complicated course, and if it is as convenience as blood donation, they would be very glad to become a donor. We reckon the low increase in donation caused by psychological reasons as conception bottleneck.

The Allocation of Additional Resources

We know the bottlenecks of transplant network are the main factors which cut down the efficiency of the network system. The former bottlenecks discussed before can come down to donation shortage except technical bottleneck. If more resources available, we only needs to allocate them on the bottlenecks.

To donation shortage, we can take lots of measures. For fast waiting-number increase, we can gather more financial resource to perfect the infrastructure of health center so as to raise the health condition. And this can also cut down the percentage of discarded organs. For low donation rate, we can promote the donation of citizens. Trying to increase the deceased donation is quite an efficient way. And the policy of presume consent is good for donation.

As to the low survival rate and short survival time, more financial resource should spend on the research of technological improvements.

Also, more labor force should be put on the health care.

Analysis on the Size of Network

The size of the transplant network system depends on the number of patients in the region and the type of organ transplanted. Normally, the number of patients is proportion to the region size. Some organs are quite difficult to match, so the size should be large enough to find the successful matched-pair. But some organs may be easier to match, so the size can be a little smaller.

Still, we can take kidney transplant as an example. The Equ.2 can be modified as follows:

$$P = (m - d / D) \cdot r \quad (3)$$

As analyzed before, we know the successfully matching probability for a kidney is:

$$m = 1 - (1 - p)^w \quad (4)$$

Where w is the total number on the waiting list and p is $1/10000$.

In the discussion of organs discard, we have get the average discarding rate is directly proportional to the total donation, and the proportion $s=14.04\%$. Then Equ.3 can be expressed as:

$$P = \{[1 - (1 - p)^w] - 0.1404\} \cdot r \quad (5)$$

For w is a variable which is relative to the size of region, we plot the successfully transplant probability of kidney as function of patients in waiting to analyze the influence of patients. **Figure 6** below shows the case of $p=1/10000$ and $r=77.1\%$:

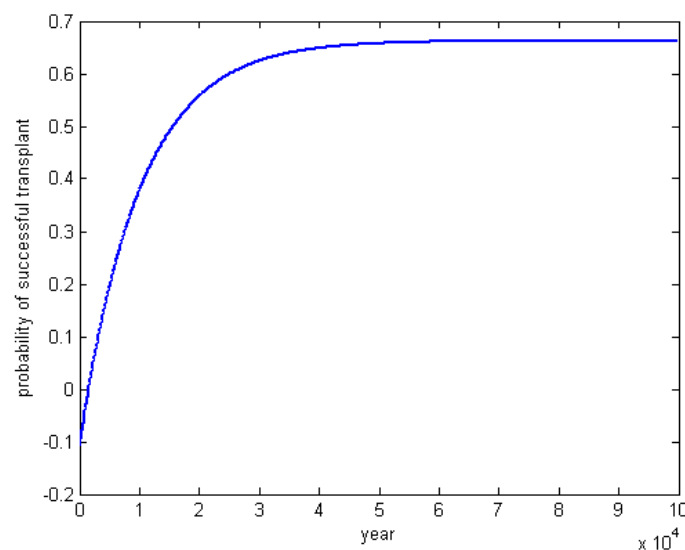


Figure 6. The trend of successfully transplant
Probability ($p=1/10000$, $r=0.771$)

From the figure we can get: the successfully utilization probability increases as the number of patients grows, and when the total patients is more than 30000, the successfully transplant probability almost reaches the maximum. In this case, we can divide large size into small ones as long as the size includes more than 30000 kidney patients.

Policies for System Efficiency

The system can be more effective by saving and prolonging more lives. If we want to save and prolong more lives, we should enlarge the number of transplant and increase the survival rate. The two factors have been taken into account in the generic model, and the increase of them can enhance the effectiveness of the transplant system. To achieve this, the practical suggestions are as follows:

In order to save more lives, we can decrease the organs discarded when the total donation remains the same. This can be achieved by taking care of the donors and trying to make use of the organ as soon as it is available.

To prolong the life expectancy, more work should be put on the improvement of transplant survival rate. Maybe to provide better medical care after organ transplant surgery is a good idea. This method quite relies on the financial resource.

Limitation of the model

We introduce a variable to represent the probability of successful match between the donor and the patient in the model. It can be used in most situations, but the value of the variable is difficult to estimate. Another limitation of this model is that it can only describe the process of transplant but not the process of donation.

Task 2: Improvement on Transplantation Policies

Investigating Other Countries' Transplant Policies

Organ transplant is a world health issue. The main problem raised is the unbalance between the donation and patients on the waiting list. Many countries make policies to raise the organ procurement and receive encouraging result.

For example, Austria implements the policy of presumed consent. Presumed consent is a principle of deceased donor. Under presumed consent legislation, a deceased individual is classified as a potential donor in absence of explicit opposition to donation before death.

After this policy implemented, in 2002 Austria has a cadavetric donation rate of 24.00 (cadavetric donors per million population), as shown in **Figure 7**. This is four times as that before [Ref. 2].

The principle of presumed consent plays an important role in recent transplant issues. Countries like Spanish, Australia, and Portugal carry out the policies successfully. The following **Figure 7** shows the world situation of organ donation divided by presumed consent and informed consent groups [Ref.3]

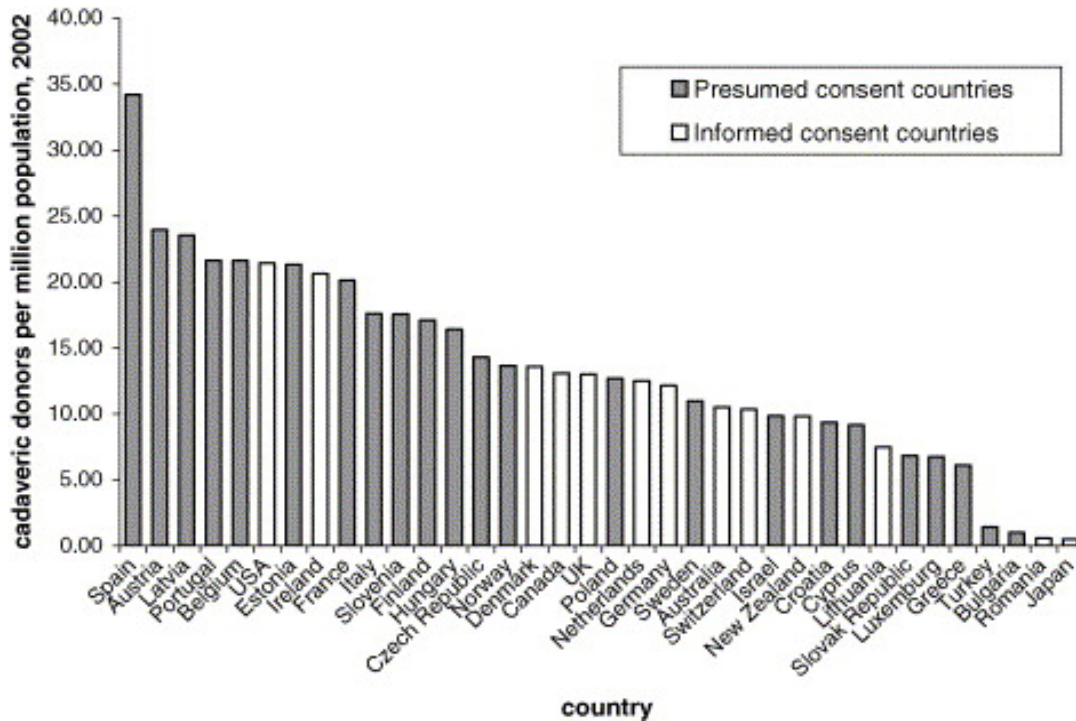


Figure 7. The cadaveric donation rate (2002).

Presumed consent countries can benefit from this principle. Statistic data shows that presumed consent countries have roughly 25–30% higher donation rates than informed consent countries on average[Ref.3].

Improvement on the model of US

The organ shortage is the main obstacle which affects the effective match of the US transplant network. And most of the organ procurements come from the voluntary donation. And figure 6 also shows now USA is an informed consent county, though actually the state of Illinois is trying to implement the policy.

Additional Assumptions

- The United States will implement the policy of presumed consent generally later.
- The increase rate is q after the policy implemented

Based on the assumptions we know the total donation will increase because of the implementation of presumed consent. If the overall increasing rate is q , the corresponding increasing rate in match i is q_i , and the donated organ in match way i is $(1+q_i)D_i$, then we modify the generic model of the successfully utilized probability and get:

$$\left\{ \begin{array}{l} P_u = \frac{\sum_{i=1}^n (D_i(1+q_i)m_i - d_i) \cdot r_i}{(1+q) \sum_{i=1}^n D_i} \\ d_i = s \cdot (1+q) \sum_{i=1}^n D_i \\ (1+q) \sum_{i=1}^n D_i = \sum_{i=1}^n D_i q_i \end{array} \right. \quad (6)$$

Where s is the portion of discarded organs in total donation, q is the average raising rate in organ donation, and q_i is the corresponding increase rate for match way i .

We calculate the successfully utilized probability of kidney again with zero-mismatches in US is:

$$P_u = \frac{[D(1+q)m - s \cdot (1+q)D] \cdot r}{D(1+q)} = (m-s) \cdot r \quad (7)$$

For kidney transplant with strict policy, the successfully utilized probability remains the same as analyzed before. But the number of patients can be saved is much larger due to the increase in total donation. As analyzes before the average in creasing rate of donation for presumed consent policy is 25%-30%. If we assume $q=27.5\%$, there will be $0.275 \times 0.6455 \times D = 0.1775D$ more patients saved with a five years survival time, where D is the total donation.

Report to the Congress

The gap between requirement and supply of organs is rising in the US. Since 1995, the number on the waiting list has grow gradually with an average growth rate of 1602 persons per year, while the organ donation is only 608 on average-just three eighths of waiting list. This means more than half of the patients could not get the chance of recovery.

This US transplant network has turned out to be ineffective and the network already indicates several bottlenecks, such as donation, concept, technical, money, etc. Many reasons account for this situation. One factor which influences the total donation is the complexity of organ donation procedure. And a portion of the organs are discarded for unpredictable risks. Besides, the low survival rate and short survival time are responsible.

Measures have to be taken immediately to optimize the transplant network. By reducing the discarded rate and promoting donation as well as shortening the size of the network, the network can work better. Through investigating other countries' transplant policies, we have found that the principle of presumed consent is another way which can enrich the organ donation by an average rising rate of 25%-30%. Since most of the states in US now carry out the informed consent policy, we suggest that the US should implement the presumed consent policy. In this case, people are assumed as organ donors before death except they propose opposition. Then the deceased donation may have a great increase thus to meet the extreme demand and release the bottleneck for the organ matching.

This is not the only way and even though the method, the total donation is still not enough. To supplement the policy and utilize the large cadaveric resource sufficiently, we recommend school education and media publicize to announce the urgent situation of the organ demand and influence the public conception. Meanwhile, technical research should be advanced to ensure higher survival rate and longer life expectancy.

Task 3: Optimizations on kidney transplant

Why should we focus on Kidney Exchange?

In 2004, 3.7 million people in the United States were diagnosed with some type of kidney disease. Hundreds of thousands of people suffer from renal failure each year and undergo dialysis or await a kidney transplant. A transplant organ can come from the cadavers of an individual who agreed to donate organs after death or from a live donor. In the US, about 68,000 patients are waiting for a kidney from a deceased donor, while each year only 10,000 are transplanted from cadavers and 6,000 from living individuals (usually relatives of the patients).

The transplanted kidney must be of a compatible ABO blood type. The 5-year survival of the transplant is enhanced by minimizing the number of mismatches on six HLA markers in the blood. At least 2,000 would-be-donor/recipient pairs are thwarted each year because of blood-type incompatibility or poor HLA match. Other sources indicate that over 6,000 people on the current waiting list have a willing but incompatible donor. This is a significant loss to the donor population and worthy of consideration when making new policies and procedures.

How to maximize the number and quality of exchanges

In order to cure more patients by transplantation, we should consider the quantity of exchanges. For the medical reason, there should be less than 3 mismatches of HLA between the donor and the recipient. At the same time, we should consider the psychological dynamics of the situation. It is possible that the patient wouldn't endow their organs for psychological reasons.

Model

In order to express the relationship in the exchange system, we develop a graph model:

Suppose a patient and his willing donor who is incompatible to be a node in the graph. If patient A's willing donor is compatible to patient B and patient B is willing to accept the donation from the willing donor of A, we set a directed edge from A to B. The weight of this edge is the number of mismatches between the willing donor of A and patient B. **Figure 8** shows the relationship.

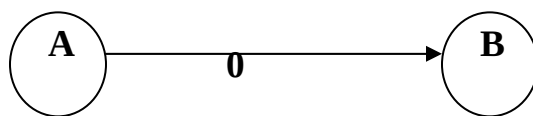


Figure 8. The relationship between patient A and B

Considering the medical factors, there shouldn't be a directed edge with a weight above 3 even if the patient is willing to accept the donation.

Now, putting all the n-way patients on the graph, we obtain a directed graph $G = \langle E, V \rangle$. If we want to make patients on the graph get kidney donation as many as possible, we should divide G into several sub-graphs which include some Hamilton graphs. Then we can optimize the network by the following

two criteria:

- a) If the total number of nodes in the Hamilton graph is at its maximum, the method of exchanging though the Hamiltonian circuit will be the best one to make most patients get donation kidney.
- b) If there is more than one Hamiltonian circuit in a sub-graph, we choose the shortest path Hamiltonian circuit, which means we can make a higher quality exchange.

Solution Algorithm

Though we have already devised a model to find the exchange method, we also need an algorithm to solve the model. In the model we developed above, medical and psychological dynamics of the situation have already been included in the topology of the graph. The purposes of our algorithm are to find the method to divide the topology into several subgraphs, and some of them are directed Hamiltonian graph. These directed Hamiltonian graphs should include as many nodes as possible, and then make the weight in the circuit as less as possible.

Since medical factor in the topology has already contain part of the quality consideration, we maximum the number of exchanges firstly, then we maximum the quality. For most of the time, we can't find the global optimal solutions, so we use greedy algorithm in our algorithm to search a locally optimal solutions.

The idiographic algorithm to look for optimal exchanges is shown below:

Step1: compute the number of nodes n in the current subgraph, let $m=0$ and $i=1$.(In this step, m is used to record the number of nodes ignored in the current subgraph; i is used to record the times we have search Hamiltonian circuit in current subgraph when the certain i nodes have been ignored.)

Step 2: If $n-m < 2$? Yes, integrate all the Hamiltonian circuit you have selected, that are the optimal exchanges, then end; else, go to step 3

Step 3: Ignore i nodes combination which haven't been ignored before in the current subgraph, search minimum Hamiltonian circuit. This will be part of the optimal exchange method. If there exist Hamiltonian circuit, go to step 5, else go to step 4 (In this step, find the minimum Hamiltonian circuit can enhance the quality of the exchange.)

Step 4: If $i = C_n^m$? Yes, $i=1$, $m=m+1$, go to step 2, else $i=i+1$, go to step 3

Step 5: Delete the nodes in the minimum Hamiltonian circuit found in step 3, obtain a new current subgraph. Go to step 1.

In step 3, we can use the DNA algorithm [Ref.4] to look for the minimum directed Hamiltonian circuit:

Step 1: Code the nodes, edges and weight of each edge in the graph

Step 2: Generate all the random paths in the graph

Step 3: Find out all the paths the start node and the end node is the same

Step 4: If there are n nodes in the graph, select the path which include the total n nodes.

Step 5: select the path pass each node only once

Step 6: Select the minimum path

The particular algorithm can be found in [Ref.4]

When the directed graph can be divided into sever Hamiltonian graph, the number of patients who can get a compatible kidney will be maximum. If all the Hamiltonian circuit is the minimum Hamiltonian circuit, the quality will reach locally maximum.

An Example of the Algorithm

A directed graph is shown in **Figure 9**.

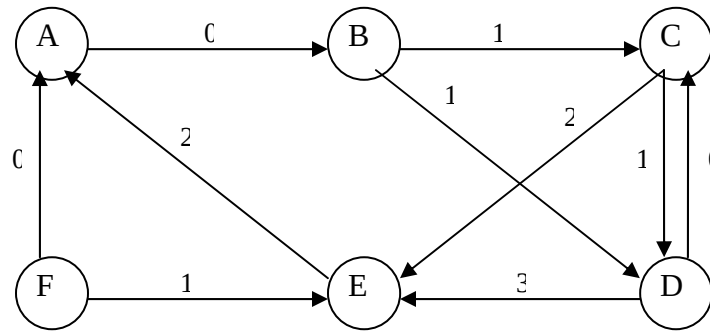


Figure 9. The directed graph to explain the algorithm

Step 1: Compute the number of nodes: $n=6$, initialize $i=1$, $m=0$;

Step 2: $n-m=6>2$, go to step 3;

Step 3: Ignore 0 nodes, there is no Hamiltonian circuit, go to step 4;

Step 4: $i = 1 = C_6^0$, $i=1$, $m=0+1=1$, go to step 2;

Step 2: $n-m=5>2$, go to step 3;

Step 3: Ignore nodes F, the minimum Hamiltonian circuit is $A \rightarrow B \rightarrow D \rightarrow C \rightarrow E \rightarrow A$, the total weight is 5. Go to step 5;

Step 5: Delete the nodes in the minimum Hamiltonian circuit found in step 3, the current graph is only the nodes F, go to step 1;

Step 1: Compute the number of nodes: $n=1$, initialize $i=1$, $m=0$;

Step 2: $n-m=1<2$, end.

The final exchange method is $A \rightarrow B \rightarrow D \rightarrow C \rightarrow E \rightarrow A$, that means the willing donor of A give his donation to patient B, B's willing donor give his donation to patient D. The same token, patient A will obtain the donation from patient E's willing donor. In this case, there will be 5 patients can get a compatible kidney while only 1 patient still can't find compatible kidney. The more people participate in the exchange system, the less probability a patient can't get a compatible kidney will be.

How Many More Annual Transplants Will Generate

Let a_k is the probability of exchanging among more than k pairs of patients and willing donors in the exchange system.

Assume that, there are n patients in the exchange system and a patient and a donor are compatible with probability p , then

$$a_{k-1} = K_{k-1}^{k-1} C_n^{k-1} [1 - (1-p)^{k-2}] (1-a_k) \quad (8)$$

$(1-a_k)$ is the probability of exchanging among less than k pairs of patients and willing donors in the exchange system, $[1 - (1-p)^{k-2}]$ is the probability of a patient can be matched with at least one willing donor in the selected $k-1$ pairs of donors and patients, C_n^{k-1} is the combination of selecting $k-1$ patients from the exchange system, K_{k-1}^{k-1} means the $k-1$ patients can be circle arrayed in order to form the exchange circle.

For $a_n = p^n$, and the expectation of annual transplants will generate in this system is (only consider the transplants generated in a exchange circle) :

$$E = \sum_{k=2}^n k(a_k - a_{k-1}) \quad (9)$$

When $p=10^{-4}$ and $n=6000$, we use MATLAB to compute the expectation of annual transplants generated in this system, the result is: 1138. That means there will be 1138 more patients will get

compatible kidney per year, if there are 6000 patients in the exchange system.

Effect on the Waiting List

From the analysis above, we find that the exchange system can increase the rate of patients getting compatible kidney. The more patients in the exchange system, the higher probability it will be to get compatible kidney. For example, if there are 600 patients in the exchange system, applying the algorithm, we get that there are only 2.83% patients who can get compatible kidney. If there are 6000 patients in the system, there will be 18.97% patients who can get compatible kidney, and the portion reaches 44.2% if there are 60000 patients in the system.

With the exchange system, patients can get a more compatible kidney. Patients should better choose to take part in the exchange system to look for a living donation, if there is a willing donor for him.

Task 4: Strategies for Deciding to Take an Offered Organ

Patients face agonizing choices when an offered organ is available but not matches very well. To wait for a better match or a live donor may bring risks to patients, but the better benefit is quite attractive. So how does a patient make his decision? In fact, whether we should take an offered kidney is determined by the expected monetary value. The strategy can be devised by comparing the expected monetary value of different choices.

Risks that Patients Face

Possible Death Before They Get Next Donation Kidney

If a patient doesn't take the kidney available from the cadaver queue, he must wait for the next kidney available no matter he chooses to wait for a better match from the cadaver or wait for a live exchange. During the time of waiting, he may be dead of illness.

The Quality of the Next Kidney

Because no one can ensure there will be a better kidney next time. Even if they choose to wait for a better match from an exchange, they still face the possible of a worse one.

Assumptions

- If a patient discards the chance to transplant, he will wait another 3 years (the average waiting time in the US) to get another kidney from the cadaver queue.
- The quality of the next kidney is unknown before it is available
- If the patient chooses to wait for a better match from an exchange, the next kidney for him is a zero-mismatch one.
- If the patient choose to wait for a better match from the cadaver, the three matches (less than three-mismatches is available) of the next kidney has an equal probability.

Quantify the Factors in Waiting

In order to estimate the expected monetary value, we should quantify the main factors which influence the patients' choice. The main factors which the patients consider are the time they should wait before

getting next kidney, the survival rate and survival time of the transplanted kidney. The waiting time is determined by the number of donor. The survival time and survival rate are determined by the HLA match and the age of the patients:

$$T_{st} = \min[(L-A), C] \quad (10)$$

Where T_{st} is the survival time of the transplanted kidney; the L is the expected life of the patient; A is the age of the patient; C is the expected life of the transplanted kidney if the patient won't die before the transplanted kidney.

So we can use the waiting time, survival rate and survival time of the transplanted kidney to quantify the strategy. Whether to accept the available donation or discard will be determined by the age of the patients and the HLA match. The quantified results of the main factors are shown in **Table 5**:

Table 5. The quantizing of the main factors influence strategy

HLA match	4	5	6	living donor	weight
survival rate (5 years) %	75.0	75.0	77.1	p_1	w_1
survival time (year)	$\min[(L-A), C_1]$	$\min[(L-A), C_2]$	$\min[(L-A), C_3]$	$\min[(L-A), C_4]$	w_2
waiting time (year)	3	3	3	5	w_3

Because these factors have different dimensions, it is difficult to estimate the overall benefits. In order to eliminate the influence of the dimensions, we normalize the three factors we quantized. Then we obtain:

Table 6. The normalized quantizing of the main factors influence strategy

HLA match	4	5	6	living donor	weight
survival rate (% 5 years)	$75.0/R$	$75.0/R$	$77.1/R$	p_1/R	w_1
survival time (year)	$\frac{\min[(L-A), C_1]}{S}$	$\frac{\min[(L-A), C_2]}{S}$	$\frac{\min[(L-A), C_3]}{S}$	$\frac{\min[(L-A), C_4]}{S}$	w_2
waiting time (year)	$3/14$	$3/14$	$3/14$	$5/14$	w_3

Where L is the expected life of the patients, C_i ($i=1,2,3,4$) is the expected life of the transplanted kidney when HLA match numbers are 4,5,6 or obtain kidney from an exchange, $R=75+75+77.1+p_1$ and

$$S = \sum_{i=1}^4 \min[(L-A) - C_i].$$

For most of the time, w_1 and w_2 are positive, while w_3 is negative.

From [Ref.5] $C_1=C_2=10.1$, $C_3=14.5$ and $C_4=29.4$

How to determine whether to accept the kidney

Based on the quantification above, we can compute the current monetary value (CMV) if one accepts the kidney available and the expected monetary value (EMV) if one chooses to wait for a better match from the cadaver queue or from an exchange:

If the HLA match of the available kidney is 4, the monetary value is:

$$CMV_4 = w_1 \frac{75.0}{R} + w_2 \frac{\min[(L-A), C_1]}{S} \quad (11)$$

If the HLA match of the available kidney is 5, the monetary value is:

$$CMV_5 = w_1 \frac{75.0}{R} + w_2 \frac{\min[(L-A), C_2]}{S} \quad (12)$$

If the HLA match of the available kidney is 6, the monetary value is:

$$CMV_6 = w_1 \frac{77.1}{R} + w_2 \frac{\min[(L-A), C_3]}{S} \quad (13)$$

If the patient chooses to wait for a better donor from the cadaver queue, the expected monetary value is:

$$EMV_d = \frac{1}{3}(CMV_4 + \frac{3}{14}w_3) + \frac{1}{3}(CMV_5 + \frac{3}{14}w_3) + \frac{1}{3}(CMV_6 + \frac{3}{14}w_3) \quad (14)$$

If the patient chooses to wait for a better donor from an exchange, the expected monetary value is:

$$EMV_l = w_1 \frac{p_1}{R} + w_2 \frac{\min[(L-A), C_4]}{S} + \frac{5}{14}w_3 \quad (15)$$

Compare the CMV if you accept the kidney and the two EMV if you choose to wait for a better match, the largest one will be the best choice.

How to Use the Strategy

Assume that, the patient is 35 years old, his expected life is 50 years, if he transplants a kidney from an exchange, the survival rate is 85% and he weight the three factors 0.7, 0.3, -0.2. Now, he has an available kidney with 5 HLA matches, should he take the offered kidney?

Compute the CMV_5 , EMV_d and EMV_l :

$CMV_5=0.2289$, $EMV_d=0.1965$, $EMV_l=0.2097$. These data show that taking the kidney transplant immediately is the best choice. To wait for another chance will bring more disadvantages.

Limitation

This model introduces a method to evaluate the advantages of different choices. But before we use this model to determine which choice is the best, there are so many parameters we should consider, and

the weight is difficult to quantify.

Task 5: Recommendations on Kidney Transplant Criteria and Policies

About Criteria

From the ethical perspective, we advocate to add the policy that children less than 18 years old can receive higher priority than candidates on the same status if the organs turn out to match well. This goes on well with the traditional conception that children are the weak groups of society, and we have responsibility to put them on the high priority with a view to giving the younger potential recipient a chance to experience some of the life that the older potential recipient has already experienced. Meanwhile, this proposal takes the benefit into account, for the life expectancy of young people is relatively long than the olds, and younger patients who haven't suffered the effects of age and attendant medical deterioration would be favored over patients in their 50s and above. So the organ should go to the person who lives the longest and have a higher survival rate.

But we should also realize that age is not the highest priority, and other factors must take into account to decide which candidate is the most suitable one. So we have to give some weight to the ethical concerns. We suggest the following rank order of criteria give out the reasons as well (the priority decreases from top to bottom).

- a) The quality of the match. For organ transplant, the matching process is the first and most critical step. Anyone whose blood type and HLA are not matched can't take the surgery at all, let alone to be the higher priority in the list.
- b) The physical distance between organ donors and recipients. Once the first priority is satisfied, the physical distance of organ should be considered. The transplant takes place between the ones whose organ work similarly or at least can work. If the size differs a lot, the organ is unsuitable.
- c) The medical urgency, age, time on the waiting list and distance between donors and recipient are on the same level of priority, but they enjoy the different weight. Medical urgency has highest weight, for we address the importance of lives. And age enjoys the second high weight, for the expected benefit on children is larger than the olds. Waiting time is on the third high weight. This is based on the equity of health care; meanwhile to save more lives in urgent if they can wait a little longer. The distance is the last consideration. Now recovered organs are stored in a cold organ preservation solution. For heart and lung recipients, it is best to transplant the organ within six hours of organ recovery. Livers can be preserved up to 24 hours after recovery [Ref.6]. The maintaining time can satisfy the transplant time, or we can make it possible to meet the time limit. The overall decision is made by the sum of the four factors with the corresponding weight.

About Policies

To the recent situation of transplant network in the United States, organ shortage is the major bottleneck which affects the effectiveness of the system, so the policy of presumed consent may be an effective way to solve the problem. Before US, many countries have implemented the principle, such as Austria. The result is obvious with a roughly 25–30% increase. But additional work needs to be done for US if they want to advocate the policy. Because the principle means a deceased individual is classified as a potential donor in absence of explicit opposition to donation before death. So the government

should try to make the citizens treat the policy positively and persuade them to believe that they can show opposition without any burden.

Ethical dimensions

In both task3 and task 4, we ignore the ethical concerns. In task 3, we treat all candidates of the change system equally. The purpose of the procedure is to match the patients as much as possible and the allocation of the organ resource is determined by the best match routes. In task 4, we only pay attention to the psychological factors, and the choice is made through the method of Expected Monetary Value (EMV). In this way, our models can be simplified.

About Organ Selling

We don't recommend organ selling even though it is an effective way to increase the supply of organs. This disobeys the conventional concept that organ donation should be an act of altruism or gifting, motivated by the desire to do good for a needy patient, or the desire to give added meaning to the donor's death. Never desire to profit from the sale of one's body or the sale of a family member's mortal remains. If so, the society would become a market of organs, and criminals will generate.

To procure more organs, we can provide financial incentives, but not selling! Public compensation for organs is a good way. Another work can be done is to change the traditional concept of mortal remains.

Task 6: Analysis on Risks of Donor

The Influence to the donors

The potential risks play an important part in the decision of the donors. The probability of success for the recipient is rather low, donors may not willing to donate their organs for they think it is a waste of organs, because the patients may not receive a successful transplant but the organs are taken out of their bodies forever. Patients gain no benefit but donors have to bear the harm. With high survival probability for the recipient but low survival probability for the donor, donors will make their decision much more carefully and the chance for patients get the organs is slight because donors may face failure of the remaining organ or even die during transplantation. Meanwhile, the possible future health problems for the donor are also considered. The health and quality of the donors' future lives will be negatively impacted. The donors may have shorter lifetime, be vulnerable to future illness, etc. A donor doesn't want to become another patient. So if the probability of future health risks is high, donors will decrease obviously. Besides, suffer from the post-operative pain and recovery also holds back the donors' enthusiasm.

The psychological problems also affect the decision of organ donors. People may have a negative attitude to the risks of donation. Some of them get phobias and is fear to become a donor. The misinformation of the health center also stops the paces of potential organ donors which makes people become irrational to organ transplant. They get scared. Besides, fears left behind the previous experiment with surgery also prevent quite a mount of potential donors.

Discussions on n-paired Network and Modification

As analyzed in Task 3, when a patient enters a n-paired network rather than receive a direct transplant

to the relative or friend, the larger the size n is, the higher the probability for him to get the minimum mismatches organ.

The influence of these factors discourages potential donors from becoming a donor indeed. We introduce a variable a as the percentage of potential donors becoming donors, and D' as the number of potential donors, thus we get $D = aD'$. With D replaced by aD' , we adjust our model to reflect the influence of the factor.

Suggestion to increase more altruistic donors

Altruists may contribute their organs selflessly for the desperate candidates on the waiting list. The donors' high trust in the surgeons and the transplant operation will also arouse enthusiasm.

Technology improvement

By improving the transplant technology, with both of the probability of success for transplant and probability of the donor's survival rate advanced, the risk reduced and less post-operative pain, more altruistic donors will be recruited.

Post-operative care and compensation

Donors can be awarded the post-operative care, less health-insurance and early eligibility for Medicare coverage, making assure of their health and cutting down their health relative cost.

Public information and communication

Full knowledge about the organ transplant is acquired and shared for the public, so when they make decision, they can weigh the risk themselves, avoid being affected by the misinformation.

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Appendix

The program to compute the **expectation of annual transplants will generate in the exchange system**

```
p=10^(-4);
n=6000;
a=zeros(1,n);
a(n)=p^n;
for k=n:-1:2
    P=ones(1,k)*(1-(1-p)^k);
    a(k-1)=sum(P)*(1-a(k))/k;
end
E=[];E(1)=0;
for k=2:n-1
    E(k)=(a(k)-a(k+1))*k;
end
E(n)=a(n)*n;
sum(E)
```

The program to compute correlation between number of patients on the waiting list and time

```
t=[1995
1996
1997
1998
1999
2000
2001
2002
2003
2004];%year
w=[17875
18308
19047
20171
22800
23427
23445
24636
25604
28320];%waiting
y=w;x=t;
n=length(x);
```

```

ax=sum(x)/n;
ay=sum(y)/n;
lxx=sum(x.^2)-n*ax^2;
lyy=sum(y.^2)-n*ay^2;
lxy=sum(x.*y)-n*ax*ay;
Rw=lxy/(sqrt(lxx)*sqrt(lyy))%correlation of kidey waitinglist and year
bw=lxy/lxx %rate

```

The program to compute correlation between number of donors and time

```

t=[1995
1996
1997
1998
1999
2000
2001
2002
2003
2004];%year
D0=[12972 12226 11878 11566 10982 10110 9760 9016 8714 8393]';
D=zeros(10,1);
for i=1:10
    D(i)=D0(11-i);%donation
end
y=D;x=t;
n=length(x);
ax=sum(x)/n;
ay=sum(y)/n;
lxx=sum(x.^2)-n*ax^2;
lyy=sum(y.^2)-n*ay^2;
lxy=sum(x.*y)-n*ax*ay;
RD=lxy/(sqrt(lxx)*sqrt(lyy))%correlation of donation and year
bD=lxy/lxx %rate
plot(t,w,t,D)

```

The program to compute the kidney discards rate

```

t=[1995
1996
1997
1998
1999
2000
2001

```

```
2002
2003
2004];%year
d=[566  557 20
623 580 28
532 670 35
545 724 38
591 859 26
675 910 48
695 816 68
613 830 59
702 878 92
835 1077 171];%kidney discards
d=sum(d,2);
rd=d./D;%rate of kidney discards
figure,plot(t,rd)
hold on,axis([1995,2004,0,1]),ylabel('kidney discards rate')
ard=sum(rd)/10
```

The program to plot the Figure 5

```
W=1:100:100000;
P=1/10000;
R=0.771;
dr=0.1404;
Y=(1-(1-P).^W-dr)*R;
figure,plot(W,Y)
hold on,ylabel('probability of successful transplant')
```